∑ Global Education Formula Handbook (Class 1 to PhD)

Comprehensive All-Level Multi-Disciplinary Formula & Scientific Reference • 580+ Core Formulas

CLASS 1 TO PHD

1. math (14 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Physics	$Class6-10, Class9-12, Class11-12, Class12-University, University-PhD$	`Class 6-10, Class 9-12, Class 11-12, Class 12-University, University-PhD`
Chemistry	$Class8-10, Class10-12, Class11-12, University-PhD$	`Class 8-10, Class 10-12, Class 11-12, University-PhD`
Biology	$Class6-12, Class11-University, University-PhD$	`Class 6-12, Class 11-University, University-PhD`
Statistics & Data Science	$Class9-12, University-PhD$	`Class 9-12, University-PhD`
Computer Science	$Class6-12, University-PhD$	`Class 6-12, University-PhD`
Engineering	$University$	`University`
Economics & Finance	$Class9-12, University-PhD$	`Class 9-12, University-PhD`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Earth & Environmental Science	$Class6-12, University-PhD$	`Class 6-12, University-PhD`
Health Sciences	$University$	`University`
Psychology & Social Science	$University-PhD$	`University-PhD`
Business, Accounting & Operations	$University$	`University`
Astronomy & Astrophysics	$University-PhD$	`University-PhD`
Reference	$AllLevels$	`All Levels`
Notation note: Formula conventions can differ between countries, curricula, and disciplines. Always check the	$definition, units, assumptions, andsignconventionbeforeusingaformulainagradedorprofessionalsetting.$	`definition, units, assumptions, and sign convention before using a formula in a graded or professional setting.`

2. Mathematics (196 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Place value 1.1 Arithmetic & Number Sense (Class 1	$Number = \sum (digit \times place value)$	`Number = \sum (digit \times place value)`
Addition 1.1 Arithmetic & Number Sense (Class 1	$a + b = b + a$	`a + b = b + a`
Associativity 1.1 Arithmetic & Number Sense (Class 1	$(a + b) + c = a + (b + c)$	`(a + b) + c = a + (b + c)`
Subtraction 1.1 Arithmetic & Number Sense (Class 1	$a - b = a + (-b)$	`a - b = a + (-b)`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Multiplication 1.1 Arithmetic & Number Sense (Class 1	$a \times b = b \times a$	`a \times b = b \times a`
Associativity 1.1 Arithmetic & Number Sense (Class 1	$(ab)c = a(bc)$	`(ab)c = a(bc)`
Distributive law 1.1 Arithmetic & Number Sense (Class 1	$a(b + c) = ab + ac$	`a(b + c) = ab + ac`
Division 1.1 Arithmetic & Number Sense (Class 1	$Dividend = Divisor \times Quotient + Remainder$	`Dividend = Divisor \times Quotient + Remainder`
Average 1.1 Arithmetic & Number Sense (Class 1	$Average = Sum of observations / Number of observations$	`Average = Sum of observations / Number of observations`
Fraction addition 1.1 Arithmetic & Number Sense (Class 1	$a/b + c/d = (ad + bc)/(bd)$	`a/b + c/d = (ad + bc)/(bd)`
Fraction subtraction 1.1 Arithmetic & Number Sense (Class 1	$a/b - c/d = (ad - bc)/(bd)$	`a/b - c/d = (ad - bc)/(bd)`
Fraction multiplication 1.1 Arithmetic & Number Sense (Class 1	$(a/b)(c/d) = ac/bd$	`(a/b)(c/d) = ac/bd`
Fraction division 1.1 Arithmetic & Number Sense (Class 1	$(a/b) \div (c/d) = ad/bc$	`(a/b) \div (c/d) = ad/bc`
Percentage 1.1 Arithmetic & Number Sense (Class 1	$Percentage = (Part / Whole) \times 100$	`Percentage = (Part / Whole) \times 100`
Unitary method 1.1 Arithmetic & Number Sense (Class 1	$Value of one unit = Total value / Number of units$	`Value of one unit = Total value / Number of units`
Ratio 1.1 Arithmetic & Number Sense (Class 1	$a:b = a/b$	`a:b = a/b`
Proportion 1.1 Arithmetic & Number Sense (Class 1	$a:b = c:d \Rightarrow ad = bc$	`a:b = c:d \Rightarrow ad = bc`
Speed 1.1 Arithmetic & Number Sense (Class 1	$Speed = Distance / Time$	`Speed = Distance / Time`
Distance 1.1 Arithmetic & Number Sense (Class 1	$Distance = Speed \times Time$	`Distance = Speed \times Time`
Time 1.1 Arithmetic & Number Sense (Class 1	$Time = Distance / Speed$	`Time = Distance / Speed`
Perimeter of square 1.2 Geometry, Measurement & Mensuration (Class 1	$P = 4a$	`P = 4a`
Area of square 1.2 Geometry, Measurement & Mensuration (Class 1	$A = a^2$	`A = a^2`
Perimeter of rectangle 1.2 Geometry, Measurement & Mensuration (Class 1	$P = 2(l + b)$	`P = 2(l + b)`
Area of rectangle 1.2 Geometry, Measurement & Mensuration (Class 1	$A = lb$	`A = lb`

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Triangle area 1.2 Geometry, Measurement & Mensuration (Class 1	$A = \frac{1}{2}bh$	`A = \frac{1}{2}bh`
Parallelogram area 1.2 Geometry, Measurement & Mensuration (Class 1	$A = bh$	`A = bh`
Circle circumference 1.2 Geometry, Measurement & Mensuration (Class 1	$C = 2\pi r = \pi d$	`C = 2\pi r = \pi d`
Circle area 1.2 Geometry, Measurement & Mensuration (Class 1	$A = \pi r^2$	`A = \pi r^2`
Cuboid volume 1.2 Geometry, Measurement & Mensuration (Class 1	$V = lbh$	`V = lbh`
Cube volume 1.2 Geometry, Measurement & Mensuration (Class 1	$V = a^3$	`V = a^3`
Cube surface area 1.2 Geometry, Measurement & Mensuration (Class 1	$TSA = 6a^2$	`TSA = 6a^2`
Cuboid surface area 1.2 Geometry, Measurement & Mensuration (Class 1	$TSA = 2(lb + bh + hl)$	`TSA = 2(lb + bh + hl)`
Temperature conversion 1.2 Geometry, Measurement & Mensuration (Class 1	$^\circ F = (9/5)^\circ C + 32$	`°F = (9/5)°C + 32`
Temperature conversion 1.2 Geometry, Measurement & Mensuration (Class 1	$^\circ C = (5/9)(^\circ F - 32)$	`°C = (5/9)(°F - 32)`
Identity 1.3 Algebra & Exponents (Class 6	$(a+b)^2 = a^2 + 2ab + b^2$	`(a + b)^2 = a^2 + 2ab + b^2`
Identity 1.3 Algebra & Exponents (Class 6	$(a-b)^2 = a^2 - 2ab + b^2$	`(a - b)^2 = a^2 - 2ab + b^2`
Difference of squares 1.3 Algebra & Exponents (Class 6	$a^2 - b^2 = (a-b)(a+b)$	`a^2 - b^2 = (a - b)(a + b)`
Cube sum 1.3 Algebra & Exponents (Class 6	$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$	`a^3 + b^3 = (a + b)(a^2 - ab + b^2)`
Cube difference 1.3 Algebra & Exponents (Class 6	$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$	`a^3 - b^3 = (a - b)(a^2 + ab + b^2)`
Three-term square 1.3 Algebra & Exponents (Class 6	$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$	`(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca`
Exponent product 1.3 Algebra & Exponents (Class 6	$a^m a^n = a^(m+n)$	`a^m a^n = a^(m+n)`
Exponent quotient 1.3 Algebra & Exponents (Class 6	$a^m/a^n = a^(m-n)$	`a^m/a^n = a^(m-n)`
Power of a power 1.3 Algebra & Exponents (Class 6	$(a^m)^n = a^(mn)$	`(a^m)^n = a^(mn)`

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Zero exponent 1.3 Algebra & Exponents (Class 6	$a^0 = 1, a \neq 0$	`aᵒ = 1, a \neq 0`
Negative exponent 1.3 Algebra & Exponents (Class 6	$a^{(} - n) = 1/a^n$	`a^(-n) = 1/a^n`
Direct proportion 1.3 Algebra & Exponents (Class 6	$y = kx$	`y = kx`
Inverse proportion 1.3 Algebra & Exponents (Class 6	$y = k/x$	`y = k/x`
Profit 1.4 Commercial Arithmetic (Class 6	$Profit = SP - CP$	`Profit = SP - CP`
Loss 1.4 Commercial Arithmetic (Class 6	$Loss = CP - SP$	`Loss = CP - SP`
Profit percent 1.4 Commercial Arithmetic (Class 6	$Profit\% = (Profit/CP) \times 100$	`Profit\% = (Profit/CP) \times 100`
Loss percent 1.4 Commercial Arithmetic (Class 6	$Loss\% = (Loss/CP) \times 100$	`Loss\% = (Loss/CP) \times 100`
Discount 1.4 Commercial Arithmetic (Class 6	$Discount = MP - SP$	`Discount = MP - SP`
Discount percent 1.4 Commercial Arithmetic (Class 6	$Discount\% = (Discount/MP) \times 100$	`Discount\% = (Discount/MP) \times 100`
Simple interest 1.4 Commercial Arithmetic (Class 6	$SI = PRT/100$	`SI = PRT/100`
Amount (SI) 1.4 Commercial Arithmetic (Class 6	$A = P + SI$	`A = P + SI`
Compound amount 1.4 Commercial Arithmetic (Class 6	$A = P(1 + r/100)^n$	`A = P(1 + r/100)^n`
Compound interest 1.4 Commercial Arithmetic (Class 6	$CI = A - P$	`CI = A - P`
Depreciation 1.4 Commercial Arithmetic (Class 6	$V = P(1 - r/100)^n$	`V = P(1 - r/100)^n`
GST/tax 1.4 Commercial Arithmetic (Class 6	$Tax = Taxrate \times Taxablevalue$	`Tax = Tax rate \times Taxable value`
Speed average 1.4 Commercial Arithmetic (Class 6	$Averagespeed = Totaldistance/Totaltime$	`Average speed = Total distance / Total time`
Distance formula 1.5 Coordinate Geometry & Triangles (Class 9	$d = sqrt[(x_2 - x_1)^2 + (y_2 - y_1)^2]$	`d = \\sqrt[(x₂-x₁)^2 + (y₂-y₁)^2]`
Midpoint 1.5 Coordinate Geometry & Triangles (Class 9	$M = ((x_1 + x_2)/2, (y_1 + y_2)/2)$	`M = ((x₁+x₂)/2, (y₁+y₂)/2)`
Section formula 1.5 Coordinate Geometry & Triangles (Class 9	$P = ((mx_2 + nx_1)/(m + n), (my_2 + ny_1)/(m + n))$	`P = ((mx₂+nx₁)/(m+n), (my₂+ny₁)/(m+n))`

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Slope 1.5 Coordinate Geometry & Triangles (Class 9	$m = (y_2 - y_1)/(x_2 - x_1)$	`m = (y₂-y₁)/(x₂-x₁)`
Line equation 1.5 Coordinate Geometry & Triangles (Class 9	$y - y_1 = m(x - x_1)$	`y - y₁ = m(x - x₁)`
Two-point line 1.5 Coordinate Geometry & Triangles (Class 9	$(y - y_1)/(y_2 - y_1) = (x - x_1)/(x_2 - x_1)$	`(y-y₁)/(y₂-y₁) = (x-x₁)/(x₂-x₁)`
Pythagorean theorem 1.5 Coordinate Geometry & Triangles (Class 9	$c^2 = a^2 + b^2$	`c^2 = a^2 + b^2`
Triangle area 1.5 Coordinate Geometry & Triangles (Class 9	$A = \frac{1}{2} \times base \times height$	`A = \frac{1}{2} \times base \times height`
Heron's formula 1.5 Coordinate Geometry & Triangles (Class 9	$s = (a + b + c)/2; A = sqrt[s(s - a)(s - b)(s - c)]$	`s = (a+b+c)/2; A = \\sqrt[s(s-a)(s-b)(s-c)]`
Equilateral triangle 1.5 Coordinate Geometry & Triangles (Class 9	$A = (\sqrt{3}/4)a^2$	`A = (\sqrt3/4)a^2`
Equilateral height 1.5 Coordinate Geometry & Triangles (Class 9	$h = (\sqrt{3}/2)a$	`h = (\sqrt3/2)a`
Arc length 1.6 Circle & Mensuration (Class 9	$L = (\theta/360°) \times 2\pi r$	`L = (\theta /360°) \times 2\pi r`
Sector area 1.6 Circle & Mensuration (Class 9	$A = (\theta/360°) \times \pi r^2$	`A = (\theta /360°) \times \pi r^2`
Ring/annulus area 1.6 Circle & Mensuration (Class 9	$A = \pi(R^2 - r^2)$	`A = \pi (R^2-r^2)`
Cylinder volume 1.6 Circle & Mensuration (Class 9	$V = \pi r^2 h$	`V = \pi r^2h`
Cylinder curved area 1.6 Circle & Mensuration (Class 9	$CSA = 2\pi rh$	`CSA = 2\pi rh`
Cylinder total area 1.6 Circle & Mensuration (Class 9	$TSA = 2\pi r(h + r)$	`TSA = 2\pi r(h+r)`
Cone volume 1.6 Circle & Mensuration (Class 9	$V = (1/3)\pi r^2 h$	`V = (1/3)\pi r^2h`
Cone slant height 1.6 Circle & Mensuration (Class 9	$l = \sqrt{(r^2 + h^2)}$	`l = \sqrt(r^2+h^2)`
Cone curved area 1.6 Circle & Mensuration (Class 9	$CSA = \pi rl$	`CSA = \pi rl`
Cone total area 1.6 Circle & Mensuration (Class 9	$TSA = \pi r(l + r)$	`TSA = \pi r(l+r)`
Sphere volume 1.6 Circle & Mensuration (Class 9	$V = (4/3)\pi r^3$	`V = (4/3)\pi r^3`

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Sphere area 1.6 Circle & Mensuration (Class 9	$A = 4\pi r^2$	`A = 4\pi r^2`
Hemisphere volume 1.6 Circle & Mensuration (Class 9	$V = (2/3)\pi r^3$	`V = (2/3)\pi r^3`
Hemisphere curved area 1.6 Circle & Mensuration (Class 9	$CSA = 2\pi r^2$	`CSA = 2\pi r^2`
Hemisphere total area 1.6 Circle & Mensuration (Class 9	$TSA = 3\pi r^2$	`TSA = 3\pi r^2`
Frustum volume 1.6 Circle & Mensuration (Class 9	$V = (1/3)\pi h(R^2 + r^2 + Rr)$	`V = (1/3)\pi h(R^2 + r^2 + Rr)`
Frustum slant height 1.6 Circle & Mensuration (Class 9	$l =$ $sqrt[h^2 + (R-r)^2]$	`l = \\sqrt[h^2 + (R-r)^2]`
Frustum curved area 1.6 Circle & Mensuration (Class 9	$CSA = \pi(R+r)l$	`CSA = \pi (R+r)l`
Quadratic roots 1.7 Quadratic, Sequences & Binomial (Class 11	$x = [-b \pm \sqrt{(b^2 - 4ac)}]/(2a)$	`x = [-b \pm \sqrt(b^2-4ac)]/(2a)`
Discriminant 1.7 Quadratic, Sequences & Binomial (Class 11	$D = b^2 - 4ac$	`D = b^2 - 4ac`
Sum of roots 1.7 Quadratic, Sequences & Binomial (Class 11	$\alpha + \beta = -b/a$	`\alpha + \beta = -b/a`
Product of roots 1.7 Quadratic, Sequences & Binomial (Class 11	$\alpha\beta = c/a$	`\alpha \beta = c/a`
AP nth term 1.7 Quadratic, Sequences & Binomial (Class 11	$a_n = a + (n-1)d$	`a_n = a + (n-1)d`
AP sum 1.7 Quadratic, Sequences & Binomial (Class 11	$S_n = n/2[2a + (n-1)d]$	`S_n = n/2 [2a + (n-1)d]`
AP sum using last term 1.7 Quadratic, Sequences & Binomial (Class 11	$S_n = n(a+l)/2$	`S_n = n(a+l)/2`
GP nth term 1.7 Quadratic, Sequences & Binomial (Class 11	$a_n = ar^(n-1)$	`a_n = ar^(n-1)`
GP sum 1.7 Quadratic, Sequences & Binomial (Class 11	$S_n = a(r^n - 1)/(r-1), r \neq 1$	`S_n = a(r^n-1)/(r-1), r\neq 1`
Infinite GP 1.7 Quadratic, Sequences & Binomial (Class 11	$S_\infty = a/(1-r), \lvert r \rvert < 1$	`S_\infty = a/(1-r), |r|<1`
Binomial theorem 1.7 Quadratic, Sequences & Binomial (Class 11	$(x+y)^n = \sum C(n,k)x^(n-k)y^k$	`(x+y)^n = \sum C(n,k)x^(n-k)y^k`
Binomial coefficient 1.7 Quadratic, Sequences & Binomial (Class 11	$C(n,r) = n!/[r!(n-r)!]$	`C(n,r) = n!/[r!(n-r)!]`

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Factorial 1.7 Quadratic, Sequences & Binomial (Class 11	$n! = n(n-1)(n-2)\ldots1$	`n! = n(n-1)(n-2)…1`
Permutation 1.7 Quadratic, Sequences & Binomial (Class 11	$nP_r = n!/(n-r)!$	`nP_r = n!/(n-r)!`
Combination 1.7 Quadratic, Sequences & Binomial (Class 11	$nC_r = n!/[r!(n-r)!]$	`nC_r = n!/[r!(n-r)!]`
Pythagorean identity 1.8 Trigonometry (Class 11	$sin^2\theta + cos^2\theta = 1$	`sin^2\theta + cos^2\theta = 1`
Tangent identity 1.8 Trigonometry (Class 11	$1 + tan^2\theta = sec^2\theta$	`1 + tan^2\theta = sec^2\theta`
Cotangent identity 1.8 Trigonometry (Class 11	$1 + cot^2\theta = csc^2\theta$	`1 + cot^2\theta = csc^2\theta`
Tangent 1.8 Trigonometry (Class 11	$tan\theta = sin\theta/cos\theta$	`tan\theta = sin\theta /cos\theta`
Cotangent 1.8 Trigonometry (Class 11	$cot\theta = cos\theta/sin\theta$	`cot\theta = cos\theta /sin\theta`
Secant 1.8 Trigonometry (Class 11	$sec\theta = 1/cos\theta$	`sec\theta = 1/cos\theta`
Cosecant 1.8 Trigonometry (Class 11	$csc\theta = 1/sin\theta$	`csc\theta = 1/sin\theta`
Sine sum 1.8 Trigonometry (Class 11	$sin(A+B) = sinAcosB + cosAsinB$	`sin(A+B)=sinA cosB + cosA sinB`
Sine difference 1.8 Trigonometry (Class 11	$sin(A-B) = sinAcosB - cosAsinB$	`sin(A-B)=sinA cosB - cosA sinB`
Cosine sum 1.8 Trigonometry (Class 11	$cos(A+B) = cosAcosB - sinAsinB$	`cos(A+B)=cosA cosB - sinA sinB`
Cosine difference 1.8 Trigonometry (Class 11	$cos(A-B) = cosAcosB + sinAsinB$	`cos(A-B)=cosA cosB + sinA sinB`
Tangent sum 1.8 Trigonometry (Class 11	$tan(A+B) = (tanA + tanB)/(1 - tanAtanB)$	`tan(A+B)=(tanA+tanB)/(1-tanA tanB)`
Double sine 1.8 Trigonometry (Class 11	$sin2A = 2sinAcosA$	`sin2A = 2sinA cosA`
Double cosine 1.8 Trigonometry (Class 11	$cos2A = cos^2A - sin^2A = 1 - 2sin^2A = 2cos^2A - 1$	`cos2A = cos^2A-sin^2A = 1-2sin^2A = 2cos^2A-1`
Double tangent 1.8 Trigonometry (Class 11	$tan2A = 2tanA/(1 - tan^2A)$	`tan2A = 2tanA/(1-tan^2A)`
Law of sines 1.8 Trigonometry (Class 11	$a/sinA = b/sinB = c/sinC = 2R$	`a/sinA = b/sinB = c/sinC = 2R`
Law of cosines 1.8 Trigonometry (Class 11	$c^2 = a^2 + b^2 - 2abcosC$	`c^2 = a^2+b^2-2ab cosC`

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Triangle area (trig) 1.8 Trigonometry (Class 11	$A = \frac{1}{2}absinC$	`A = \frac{1}{2}ab sinC`
Power rule 1.9 Differential & Integral Calculus (Class 11	$d(x^n)/dx = nx^{(}n-1)$	`d(x^n)/dx = nx^(n-1)`
Constant derivative 1.9 Differential & Integral Calculus (Class 11	$d(c)/dx = 0$	`d(c)/dx = 0`
Exponential 1.9 Differential & Integral Calculus (Class 11	$d(e^x)/dx = e^x$	`d(e^x)/dx = e^x`
General exponential 1.9 Differential & Integral Calculus (Class 11	$d(a^x)/dx = a^x lna$	`d(a^x)/dx = a^x ln a`
Logarithm 1.9 Differential & Integral Calculus (Class 11	$d(lnx)/dx = 1/x$	`d(ln x)/dx = 1/x`
Sine 1.9 Differential & Integral Calculus (Class 11	$d(sinx)/dx = cosx$	`d(sin x)/dx = cos x`
Cosine 1.9 Differential & Integral Calculus (Class 11	$d(cosx)/dx = -sinx$	`d(cos x)/dx = -sin x`
Tangent 1.9 Differential & Integral Calculus (Class 11	$d(tanx)/dx = sec^2x$	`d(tan x)/dx = sec^2x`
Product rule 1.9 Differential & Integral Calculus (Class 11	$(fg)' = f'g + fg'$	`(fg)' = f'g + fg'`
Quotient rule 1.9 Differential & Integral Calculus (Class 11	$(f/g)' = (f'g - fg')/g^2$	`(f/g)' = (f'g-fg')/g^2`
Chain rule 1.9 Differential & Integral Calculus (Class 11	$df(g(x))/dx = f'(g(x))g'(x)$	`d f(g(x))/dx = f'(g(x))g'(x)`
Integral power 1.9 Differential & Integral Calculus (Class 11	$\int x^n dx = x^{(}n+1)/(n+1) + C, n \neq -1$	`\int x^n dx = x^(n+1)/(n+1)+C, n\neq -1`
Integral reciprocal 1.9 Differential & Integral Calculus (Class 11	$\int dx/x = ln\vert x\vert + C$	`\int dx/x = ln|x| + C`
Exponential integral 1.9 Differential & Integral Calculus (Class 11	$\int e^x dx = e^x + C$	`\int e^x dx = e^x + C`
Sine integral 1.9 Differential & Integral Calculus (Class 11	$\int sinxdx = -cosx + C$	`\int sin x dx = -cos x + C`
Cosine integral 1.9 Differential & Integral Calculus (Class 11	$\int cosxdx = sinx + C$	`\int cos x dx = sin x + C`

<table>
<tr><th>CONCEPT / FORMULA NAME</th><th>RENDERED MATHEMATICAL OUTPUT</th><th>LATEX CODE</th></tr>
<tr><td>Tangent integral
1.9 Differential & Integral Calculus (Class 11</td><td>$\int tanxdx = -ln|cosx| + C$</td><td>\int tan x dx = -ln|cos x| + C</td></tr>
<tr><td>Integration by parts
1.9 Differential & Integral Calculus (Class 11</td><td>$\int udv = uv - \int vdu$</td><td>\int u dv = uv - \int v du</td></tr>
<tr><td>Fundamental theorem
1.9 Differential & Integral Calculus (Class 11</td><td>$\int_a^b f(x)dx = F(b) - F(a), whereF' = f$</td><td>\int _a^b f(x)dx = F(b)-F(a), where F'=f</td></tr>
<tr><td>Average value
1.9 Differential & Integral Calculus (Class 11</td><td>$f_avg = (1/(b-a))\int_a^b f(x)dx$</td><td>f_avg = (1/(b-a))\int _a^b f(x)dx</td></tr>
<tr><td>Mean
1.10 Probability & Statistics (Class 11</td><td>$\bar{x} = \sum x/n$</td><td>\bar{x} = \sum x/n</td></tr>
<tr><td>Weighted mean
1.10 Probability & Statistics (Class 11</td><td>$\bar{x}_w = \sum w_ix_i / \sum w_i$</td><td>\bar{x}_w = \sum w_ix_i / \sum w_i</td></tr>
<tr><td>Variance (population)
1.10 Probability & Statistics (Class 11</td><td>$\sigma^2 = \sum(x-\mu)^2/N$</td><td>\sigma ^2 = \sum (x-\mu)^2/N</td></tr>
<tr><td>Variance (sample)
1.10 Probability & Statistics (Class 11</td><td>$s^2 = \sum(x-\bar{x})^2/(n-1)$</td><td>s^2 = \sum (x-\bar{x})^2/(n-1)</td></tr>
<tr><td>Standard deviation
1.10 Probability & Statistics (Class 11</td><td>$sigma = textsqrtvariance$</td><td>\\sigma = \\text{sqrtvariance}</td></tr>
<tr><td>Probability complement
1.10 Probability & Statistics (Class 11</td><td>$P(A^c) = 1 - P(A)$</td><td>P(Aᶜ)=1-P(A)</td></tr>
<tr><td>Addition rule
1.10 Probability & Statistics (Class 11</td><td>$P(A \cup B) = P(A) + P(B) - P(A \cap B)$</td><td>P(A∪B)=P(A)+P(B)-P(A∩B)</td></tr>
<tr><td>Conditional probability
1.10 Probability & Statistics (Class 11</td><td>$P(A|B) = P(A \cap B)/P(B)$</td><td>P(A|B)=P(A∩B)/P(B)</td></tr>
<tr><td>Multiplication rule
1.10 Probability & Statistics (Class 11</td><td>$P(A \cap B) = P(A|B)P(B)$</td><td>P(A∩B)=P(A|B)P(B)</td></tr>
<tr><td>Independent events
1.10 Probability & Statistics (Class 11</td><td>$P(A \cap B) = P(A)P(B)$</td><td>P(A∩B)=P(A)P(B)</td></tr>
<tr><td>Bayes theorem
1.10 Probability & Statistics (Class 11</td><td>$P(A|B) = P(B|A)P(A)/P(B)$</td><td>P(A|B)=P(B|A)P(A)/P(B)</td></tr>
<tr><td>Binomial probability
1.10 Probability & Statistics (Class 11</td><td>$P(X = k) = C(n,k)p^k(1-p)^(n-k)$</td><td>P(X=k)=C(n,k)p^k(1-p)^(n-k)</td></tr>
<tr><td>Binomial mean
1.10 Probability & Statistics (Class 11</td><td>$\mu = np$</td><td>\mu =np</td></tr>
<tr><td>Binomial variance
1.10 Probability & Statistics (Class 11</td><td>$\sigma^2 = np(1-p)$</td><td>\sigma ^2=np(1-p)</td></tr>
</table>

<table>
<thead>
<tr><th>CONCEPT / FORMULA NAME</th><th>RENDERED MATHEMATICAL OUTPUT</th><th>LATEX CODE</th></tr>
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<tbody>
<tr><td>Poisson probability
1.10 Probability & Statistics (Class 11</td><td>$P(X = k) = e^{(}-\lambda)\lambda^k/k!$</td><td>P(X=k)=e^(-\lambda)\lambda ^k/k!</td></tr>
<tr><td>Poisson mean/variance
1.10 Probability & Statistics (Class 11</td><td>$\mu = \sigma^2 = \lambda$</td><td>\mu =\sigma ^2=\lambda</td></tr>
<tr><td>Normal z-score
1.10 Probability & Statistics (Class 11</td><td>$z = (x - \mu)/\sigma$</td><td>z=(x-\mu)/\sigma</td></tr>
<tr><td>Coefficient of variation
1.10 Probability & Statistics (Class 11</td><td>$CV = (\sigma/\mu) \times 100\%$</td><td>CV=(\sigma /\mu)\times 100\%</td></tr>
<tr><td>Matrix product
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$(AB)_ij = \sum_k a_ikb_kj$</td><td>(AB)_ij = \sum _k a_ik b_kj</td></tr>
<tr><td>2×2 determinant
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$\lvert A\rvert = ad - bc$</td><td>|A| = ad-bc</td></tr>
<tr><td>2×2 inverse
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$A^{-1} = (1/(ad - bc))[[d, -b], [-c, a]]$</td><td>A⁻¹ = (1/(ad-bc)) [[d,-b], [-c,a]]</td></tr>
<tr><td>Eigenvalue equation
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$det(A - \lambda I) = 0$</td><td>det(A-\lambda I)=0</td></tr>
<tr><td>Eigenvector equation
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$Av = \lambda v$</td><td>Av=\lambda v</td></tr>
<tr><td>Trace
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$tr(A) = \sum_i a_ii = \sum eigenvalues$</td><td>tr(A)=\sum _i a_ii = \sum eigenvalues</td></tr>
<tr><td>Rank-nullity
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$rank(A) + nullity(A) = n$</td><td>rank(A)+nullity(A)=n</td></tr>
<tr><td>Dot product
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$u\cdot v = \sum u_iv_i = \lvert\lvert u\rvert\rvert\lvert\lvert v\rvert\rvert cos\theta$</td><td>u·v = \sum u_iv_i = ||u||||v||cos\theta</td></tr>
<tr><td>Cauchy–Schwarz
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$\lvert u\cdot v\rvert \leq \lvert\lvert u\rvert\rvert\lvert\lvert v\rvert\rvert$</td><td>|u·v| \leq ||u||||v||</td></tr>
<tr><td>Taylor series
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$f(x) = \sum_{n=0} \infty f^{(}n)(a)(x - a)^n/n!$</td><td>f(x)=\sum _{n=0}\infty f^(n)(a)(x-a)^n/n!</td></tr>
<tr><td>Maclaurin series
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$f(x) = \sum_{n=0} \infty f^{(}n)(0)x^n/n!$</td><td>f(x)=\sum _{n=0}\infty f^(n)(0)x^n/n!</td></tr>
<tr><td>L'Hôpital rule
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$limf/g = limf'/g'underapplicableconditions$</td><td>lim f/g = lim f'/g' under applicable conditions</td></tr>
<tr><td>Gradient
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$\nabla f = (\partial f/\partial x_1, \ldots, \partial f/\partial x_n)$</td><td>\nabla f = (\partial f/\partial x₁,…,\partial f/\partial x_n)</td></tr>
<tr><td>Divergence
1.11 Linear Algebra & Advanced Mathematics (University</td><td>$\nabla\cdot F = \sum_i \partial F_i/\partial x_i$</td><td>\nabla ·F = \sum _i \partial F_i/\partial x_i</td></tr>
</tbody>
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CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Curl 1.11 Linear Algebra & Advanced Mathematics (University	$\nabla \times F$	`\nabla \times F`
Laplacian 1.11 Linear Algebra & Advanced Mathematics (University	$\nabla^2 f = \sum_i \partial^2 f/\partial x_i^2$	`\nabla ^2f = \sum _i \partial ^2f/\partial x_i^2`
Fourier transform 1.11 Linear Algebra & Advanced Mathematics (University	$F(\omega) = \int_{-\infty} \infty f(t)e^{(} - i\omega t)dt$	`F(\omega )=\int _{-\infty }\infty f(t)e^(-i\omega t)dt`
Inverse Fourier transform 1.11 Linear Algebra & Advanced Mathematics (University	$f(t) = (1/2\pi)\int F(\omega)e^{(}i\omega t)d\omega$	`f(t)=(1/2\pi )\int F(\omega )e^(i\omega t)d\omega`
Laplace transform 1.11 Linear Algebra & Advanced Mathematics (University	$Lf(t) = \int_0 \infty e^{(} - st)f(t)dt$	`L{f(t)}=\int _0\infty e^(-st)f(t)dt`
Convolution 1.11 Linear Algebra & Advanced Mathematics (University	$(f * g)(t) = \int f(\tau)g(t-\tau)d\tau$	`(f*g)(t)=\int f(τ)g(t-τ)dτ`
Gradient 14.1 Vector Calculus, PDE & Numerical Methods (University	$\nabla f = (f_x, f_y, f_z)$	`\nabla f = (f_x,f_y,f_z)`
Directional derivative 14.1 Vector Calculus, PDE & Numerical Methods (University	$D_u f = \nabla f \cdot u, \lVert u \rVert = 1$	`D_u f = \nabla f·u, ||u||=1`
Divergence 14.1 Vector Calculus, PDE & Numerical Methods (University	$\nabla \cdot F = \partial F_x/\partial x + \partial F_y/\partial y + \partial F_z/\partial z$	`\nabla ·F = \partial F_x/\partial x + \partial F_y/\partial y + \partial F_z/\partial z`
Laplacian 14.1 Vector Calculus, PDE & Numerical Methods (University	$\nabla^2 f = f_x x + f_y y + f_z z$	`\nabla ^2f = f_xx+f_yy+f_zz`
Green theorem 14.1 Vector Calculus, PDE & Numerical Methods (University	$\oint_C (P\,dx + Qdy) = \iint_D (Q_x - P_y)dA$	`∮_C(Pdx+Qdy)=∬_D(Q_x-P_y)dA`
Divergence theorem 14.1 Vector Calculus, PDE & Numerical Methods (University	$\oiint_S F \cdot ndS = \iiint_V \nabla \cdot F\,dV$	`∯_S F·n dS = ∭_V \nabla ·F dV`
Stokes theorem 14.1 Vector Calculus, PDE & Numerical Methods (University	$\oint_C F \cdot dr = \iint_S (\nabla \times F) \cdot ndS$	`∮_C F·dr = ∬_S (\nabla \times F)·n dS`
Heat equation 14.1 Vector Calculus, PDE & Numerical Methods (University	$u_t = \alpha u_x x$	`u_t = \alpha u_xx`
Wave equation 14.1 Vector Calculus, PDE & Numerical Methods (University	$u_t t = c^2 u_x x$	`u_tt = c^2u_xx`
Laplace equation 14.1 Vector Calculus, PDE & Numerical Methods (University	$\nabla^2 u = 0$	`\nabla ^2u = 0`
Poisson equation 14.1 Vector Calculus, PDE & Numerical Methods (University	$\nabla^2 u = f$	`\nabla ^2u = f`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Euler method 14.1 Vector Calculus, PDE & Numerical Methods (University	$y_{n+1}=y_n+hf(x_n,y_n)$	`y_{n+1}=y_n+h f(x_n,y_n)`
RK4 14.1 Vector Calculus, PDE & Numerical Methods (University	$y_{n+1}=y_n+(h/6)(k_1+2k_2+2k_3+k_4)$	`y_{n+1}=y_n+(h/6)(k₁+2k₂+2k₃+k₄)`
Newton iteration 14.1 Vector Calculus, PDE & Numerical Methods (University	$x_{n+1}=x_n-f(x_n)/f'(x_n)$	`x_{n+1}=x_n-f(x_n)/f'(x_n)`
Trapezoidal rule 14.1 Vector Calculus, PDE & Numerical Methods (University	$\int_a^b f dx \approx h[(f_0+f_n)/2+\sum f_i]$	`\int _a^b f dx \approx h[(f₀+f_n)/2+\sum f_i]`
Simpson 1/3 rule 14.1 Vector Calculus, PDE & Numerical Methods (University	$\int \approx h/3[f_0+f_n+4\sum f_odd+2\sum f_even]$	`\int \approx h/3[f₀+f_n+4\sum f_odd+2\sum f_even]`

3. Physics (118 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Speed 2.1 Mechanics & Properties of Matter (Class 6	$v = s/t$	`v = s/t`
Acceleration 2.1 Mechanics & Properties of Matter (Class 6	$a = \Delta v/\Delta t$	`a = \Delta v/\Delta t`
Newton's second law 2.1 Mechanics & Properties of Matter (Class 6	$F = ma$	`F = ma`
Momentum 2.1 Mechanics & Properties of Matter (Class 6	$p = mv$	`p = mv`
Impulse 2.1 Mechanics & Properties of Matter (Class 6	$J = F\Delta t = \Delta p$	`J = F\Delta t = \Delta p`
Weight 2.1 Mechanics & Properties of Matter (Class 6	$W = mg$	`W = mg`
Work 2.1 Mechanics & Properties of Matter (Class 6	$W = Fs cos\theta$	`W = Fs cos\theta`
Kinetic energy 2.1 Mechanics & Properties of Matter (Class 6	$K = \frac{1}{2}mv^2$	`K = \frac{1}{2}mv^2`
Potential energy 2.1 Mechanics & Properties of Matter (Class 6	$U = mgh$	`U = mgh`
Power 2.1 Mechanics & Properties of Matter (Class 6	$P = W/t$	`P = W/t`
Pressure 2.1 Mechanics & Properties of Matter (Class 6	$p = F/A$	`p = F/A`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Density 2.1 Mechanics & Properties of Matter (Class 6	$\rho = m/V$	`\rho = m/V`
Relative density 2.1 Mechanics & Properties of Matter (Class 6	$RD = \rho_s ubstance/\rho_w ater$	`RD = \rho _substance/\rho _water`
Mechanical advantage 2.1 Mechanics & Properties of Matter (Class 6	$MA = Load/Effort$	`MA = Load/Effort`
Efficiency 2.1 Mechanics & Properties of Matter (Class 6	$\eta = (Usefuloutput/Input) \times 100\%$	`η = (Useful output/Input)\times 100\%`
First equation 2.2 Kinematics (Class 9	$v = u + at$	`v = u + at`
Second equation 2.2 Kinematics (Class 9	$s = ut + \frac{1}{2}at^2$	`s = ut + \frac{1}{2}at^2`
Third equation 2.2 Kinematics (Class 9	$v^2 = u^2 + 2as$	`v^2 = u^2 + 2as`
Average velocity 2.2 Kinematics (Class 9	$v_a vg = displacement/time$	`v_avg = displacement/time`
Uniform circular speed 2.2 Kinematics (Class 9	$v = \omega r$	`v = \omega r`
Angular velocity 2.2 Kinematics (Class 9	$\omega = \Delta\theta/\Delta t$	`\omega = \Delta \theta /\Delta t`
Angular acceleration 2.2 Kinematics (Class 9	$\alpha = \Delta\omega/\Delta t$	`\alpha = \Delta \omega /\Delta t`
Tangential acceleration 2.2 Kinematics (Class 9	$a_t = \alpha r$	`a_t = \alpha r`
Centripetal acceleration 2.2 Kinematics (Class 9	$a_c = v^2/r = \omega^2 r$	`a_c = v^2/r = \omega ^2r`
Centripetal force 2.2 Kinematics (Class 9	$F_c = mv^2/r = m\omega^2 r$	`F_c = mv^2/r = m\omega ^2r`
Projectile horizontal range 2.2 Kinematics (Class 9	$R = u^2 sin2\theta/g$	`R = u^2 sin2\theta /g`
Projectile time of flight 2.2 Kinematics (Class 9	$T = 2usin\theta/g$	`T = 2u sin\theta /g`
Projectile maximum height 2.2 Kinematics (Class 9	$H = u^2 sin^2\theta/(2g)$	`H = u^2sin^2\theta /(2g)`
Torque 2.3 Rotational Mechanics & Gravitation (Class 11	$\tau = r \times F; \lvert\tau\rvert = rF sin\theta$	`τ = r\times F; |τ| = rF sin\theta`
Angular momentum 2.3 Rotational Mechanics & Gravitation (Class 11	$L = I\omega$	`L = I\omega`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Rotational kinetic energy 2.3 Rotational Mechanics & Gravitation (Class 11	$K_rot = \frac{1}{2}I\omega^2$	`K_rot = \frac{1}{2}I\omega ^2`
Rotational equation 2.3 Rotational Mechanics & Gravitation (Class 11	$\tau = I\alpha$	`τ = I\alpha`
Parallel-axis theorem 2.3 Rotational Mechanics & Gravitation (Class 11	$I = I_CM + Md^2$	`I = I_CM + Md^2`
Perpendicular-axis theorem 2.3 Rotational Mechanics & Gravitation (Class 11	$I_z = I_x + I_y$	`I_z = I_x + I_y`
Gravitational force 2.3 Rotational Mechanics & Gravitation (Class 11	$F = Gm_1m_2/r^2$	`F = Gm₁m₂/r^2`
Gravitational field 2.3 Rotational Mechanics & Gravitation (Class 11	$g = GM/r^2$	`g = GM/r^2`
Potential 2.3 Rotational Mechanics & Gravitation (Class 11	$V = -GM/r$	`V = −GM/r`
Potential energy 2.3 Rotational Mechanics & Gravitation (Class 11	$U = -GMm/r$	`U = −GMm/r`
Escape speed 2.3 Rotational Mechanics & Gravitation (Class 11	$v_e = \sqrt(2GM/R) = \sqrt(2gR)$	`v_e = \sqrt(2GM/R) = \sqrt(2gR)`
Orbital speed 2.3 Rotational Mechanics & Gravitation (Class 11	$v_o = \sqrt(GM/r)$	`v_o = \sqrt(GM/r)`
Kepler third law 2.3 Rotational Mechanics & Gravitation (Class 11	$T^2 \propto r^3$	`T^2 ∝ r^3`
Continuity equation 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$A_1v_1 = A_2v_2$	`A₁v₁ = A₂v₂`
Bernoulli equation 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$P + \frac{1}{2}\rho v^2 + \rho gh = constant$	`P + \frac{1}{2}\rho v^2 + \rho gh = constant`
Hydrostatic pressure 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$P = P_0 + \rho gh$	`P = P₀ + \rho gh`
Buoyant force 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$F_B = \rho_fluid V_displaced g$	`F_B = \rho _fluid V_displaced g`
Young modulus 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$Y = stress/strain$	`Y = stress/strain`
Bulk modulus 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$K = -\Delta P/(\Delta V/V)$	`K = −\Delta P/(\Delta V/V)`
Shear modulus 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$G = shear stress/shear strain$	`G = shear stress/shear strain`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Ideal gas law 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$PV = nRT$	`PV = nRT`
Combined gas law 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$P_1V_1/T_1 = P_2V_2/T_2$	`P₁V₁/T₁ = P₂V₂/T₂`
First law 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$\Delta Q = \Delta U + W$	`\Delta Q = \Delta U + W`
Work at constant pressure 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$W = P\Delta V$	`W = P\Delta V`
Carnot efficiency 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$\eta = 1 - T_C/T_H$	`η = 1 - T_C/T_H`
Entropy differential 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$dS = \delta Q_rev/T$	`dS = \delta Q_rev/T`
Heat capacity 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$Q = mc\Delta T$	`Q = mc\Delta T`
Latent heat 2.4 Fluids, Elasticity & Thermodynamics (Class 11	$Q = mL$	`Q = mL`
Wave speed 2.5 Waves, SHM & Sound (Class 11	$v = f\lambda$	`v = f\lambda`
Angular frequency 2.5 Waves, SHM & Sound (Class 11	$\omega = 2\pi f$	`\omega = 2\pi f`
SHM displacement 2.5 Waves, SHM & Sound (Class 11	$x = Acos(\omega t + \varphi)$	`x = A cos(\omega t+ϕ)`
SHM velocity 2.5 Waves, SHM & Sound (Class 11	$v = -A\omega sin(\omega t + \varphi)$	`v = -A\omega sin(\omega t+ϕ)`
SHM acceleration 2.5 Waves, SHM & Sound (Class 11	$a = -\omega^2 x$	`a = -\omega ^2x`
Mass-spring period 2.5 Waves, SHM & Sound (Class 11	$T = 2\pi\sqrt{(m/k)}$	`T = 2\pi \sqrt(m/k)`
Simple pendulum period 2.5 Waves, SHM & Sound (Class 11	$T = 2\pi\sqrt{(L/g)}$	`T = 2\pi \sqrt(L/g)`
Wave equation 2.5 Waves, SHM & Sound (Class 11	$\partial^2 y/\partial x^2 = (1/v^2)\partial^2 y/\partial t^2$	`\partial ^2y/\partial x^2 = (1/v^2)\partial ^2y/\partial t^2`
Intensity 2.5 Waves, SHM & Sound (Class 11	$I = Power/Area$	`I = Power/Area`
Sound level 2.5 Waves, SHM & Sound (Class 11	$\beta = 10log_{10}(I/I_0)dB$	`\beta = 10 log₁₀(I/I₀) dB`
Beat frequency 2.5 Waves, SHM & Sound (Class 11	$f_b = \lvert f_1 - f_2 \rvert$	`f_b = |f₁-f₂|`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Doppler effect 2.5 Waves, SHM & Sound (Class 11	$f' = f(v \pm v_o)/(v \mp v_s)$	`f' = f(v\pm v_o)/(v\mp v_s)`
Coulomb law 2.6 Electrostatics & Circuits (Class 12	$F = (1/(4\pi \varepsilon_0))q_1q_2/r^2$	`F = (1/(4\pi \varepsilon ₀)) q₁q₂/r^2`
Electric field 2.6 Electrostatics & Circuits (Class 12	$E = F/q$	`E = F/q`
Point charge field 2.6 Electrostatics & Circuits (Class 12	$E = (1/(4\pi \varepsilon_0))q/r^2$	`E = (1/(4\pi \varepsilon ₀))q/r^2`
Electric potential 2.6 Electrostatics & Circuits (Class 12	$V = (1/(4\pi \varepsilon_0))q/r$	`V = (1/(4\pi \varepsilon ₀))q/r`
Potential energy 2.6 Electrostatics & Circuits (Class 12	$U = qV$	`U = qV`
Capacitance 2.6 Electrostatics & Circuits (Class 12	$C = Q/V$	`C = Q/V`
Parallel plate capacitor 2.6 Electrostatics & Circuits (Class 12	$C = \varepsilon A/d$	`C = \varepsilon A/d`
Capacitor energy 2.6 Electrostatics & Circuits (Class 12	$U = \frac{1}{2}CV^2 = Q^2/(2C) = \frac{1}{2}QV$	`U = \frac{1}{2}CV^2 = Q^2/(2C) = \frac{1}{2}QV`
Ohm law 2.6 Electrostatics & Circuits (Class 12	$V = IR$	`V = IR`
Resistivity 2.6 Electrostatics & Circuits (Class 12	$R = \rho L/A$	`R = \rho L/A`
Electrical power 2.6 Electrostatics & Circuits (Class 12	$P = VI = I^2R = V^2/R$	`P = VI = I^2R = V^2/R`
Kirchhoff junction 2.6 Electrostatics & Circuits (Class 12	$\sum I_in = \sum I_out$	`\sum I_in = \sum I_out`
Kirchhoff loop 2.6 Electrostatics & Circuits (Class 12	$\sum \Delta V = 0$	`\sum \Delta V = 0`
Series resistance 2.6 Electrostatics & Circuits (Class 12	$R_eq = \sum R_i$	`R_eq = \sum R_i`
Parallel resistance 2.6 Electrostatics & Circuits (Class 12	$1/R_eq = \sum (1/R_i)$	`1/R_eq = \sum (1/R_i)`
RC time constant 2.6 Electrostatics & Circuits (Class 12	$\tau = RC$	`τ = RC`
Lorentz force 2.7 Electromagnetism (University	$F = q(E + v \times B)$	`F = q(E + v\times B)`
Magnetic force 2.7 Electromagnetism (University	$F = qvB sin\theta$	`F = qvB sin\theta`

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Current force 2.7 Electromagnetism (University	$F = IL \times B$	`F = I L\times B`
Biot–Savart law 2.7 Electromagnetism (University	$dB = (\mu_0/4\pi)I(d\ell \times \hat{r})/r^2$	`dB = (\mu ₀/4\pi ) I(dℓ\times r̂)/r^2`
Ampere law 2.7 Electromagnetism (University	$\oint B\cdot d\ell = \mu_0 I_enclosed$	`∮B·dℓ = \mu ₀I_enclosed`
Faraday law 2.7 Electromagnetism (University	$\varepsilon = -d\Phi_B/dt$	`\varepsilon = -dΦ_B/dt`
Magnetic flux 2.7 Electromagnetism (University	$\Phi_B = \int B\cdot dA$	`Φ_B = \int B·dA`
Lenz law 2.7 Electromagnetism (University	$Directionofinducedemfopposesfluxchange$	`Direction of induced emf opposes flux change`
Inductor voltage 2.7 Electromagnetism (University	$V = LdI/dt$	`V = L dI/dt`
Inductor energy 2.7 Electromagnetism (University	$U = \frac{1}{2}LI^2$	`U = \frac{1}{2}LI^2`
Maxwell–Gauss electric 2.7 Electromagnetism (University	$\nabla\cdot E = \rho/\varepsilon_0$	`\nabla ·E = \rho /\varepsilon ₀`
Maxwell–Gauss magnetic 2.7 Electromagnetism (University	$\nabla\cdot B = 0$	`\nabla ·B = 0`
Maxwell–Faraday 2.7 Electromagnetism (University	$\nabla \times E = -\partial B/\partial t$	`\nabla \times E = -\partial B/\partial t`
Maxwell–Ampere 2.7 Electromagnetism (University	$\nabla \times B = \mu_0 J + \mu_0\varepsilon_0\partial E/\partial t$	`\nabla \times B = \mu ₀J + \mu ₀\varepsilon ₀\partial E/\partial t`
EM wave speed 2.7 Electromagnetism (University	$c = 1/\sqrt{(\mu_0\varepsilon_0)}$	`c = 1/\sqrt(\mu ₀\varepsilon ₀)`
Poynting vector 2.7 Electromagnetism (University	$S = (1/\mu_0)E \times B$	`S = (1/\mu ₀)E\times B`
Planck relation 2.8 Quantum, Relativity & Statistical Physics (University	$E = hv = \hbar\omega$	`E = hv = \hbar \omega`
de Broglie wavelength 2.8 Quantum, Relativity & Statistical Physics (University	$\lambda = h/p$	`\lambda = h/p`
Heisenberg uncertainty 2.8 Quantum, Relativity & Statistical Physics (University	$\Delta x\Delta p \geq \hbar/2$	`\Delta x \Delta p \geq \hbar /2`
Schrödinger equation 2.8 Quantum, Relativity & Statistical Physics (University	$i\hbar\partial\psi/\partial t = \hat{H}\psi$	`i\hbar \partial ψ/\partial t = \hat{H}ψ`
Time-independent Schrödinger 2.8 Quantum, Relativity & Statistical Physics (University	$\hat{H}\psi = E\psi$	`\hat{H}ψ = Eψ`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Probability density 2.8 Quantum, Relativity & Statistical Physics (University	$\rho = \|\psi\|^2$	`\rho = |ψ|^2`
Normalization 2.8 Quantum, Relativity & Statistical Physics (University	$\int \|\psi\|^2 d\tau = 1$	`\int |ψ|^2 dτ = 1`
Commutator 2.8 Quantum, Relativity & Statistical Physics (University	$[A, B] = AB - BA$	`[A,B] = AB-BA`
Energy-momentum 2.8 Quantum, Relativity & Statistical Physics (University	$E^2 = p^2c^2 + m^2c^4$	`E^2 = p^2c^2 + m^2c^4`
Relativistic energy 2.8 Quantum, Relativity & Statistical Physics (University	$E = \gamma mc^2$	`E = \gamma mc^2`
Lorentz factor 2.8 Quantum, Relativity & Statistical Physics (University	$\gamma = 1/\sqrt{(1 - v^2/c^2)}$	`\gamma = 1/\sqrt(1-v^2/c^2)`
Time dilation 2.8 Quantum, Relativity & Statistical Physics (University	$\Delta t = \gamma \Delta t_0$	`\Delta t = \gamma \Delta t₀`
Length contraction 2.8 Quantum, Relativity & Statistical Physics (University	$L = L_0/\gamma$	`L = L₀/\gamma`
Mass-energy rest 2.8 Quantum, Relativity & Statistical Physics (University	$E_0 = mc^2$	`E₀ = mc^2`
Boltzmann factor 2.8 Quantum, Relativity & Statistical Physics (University	$P \propto e^(- E/k_B T)$	`P ∝ e^(-E/k_BT)`
Partition function 2.8 Quantum, Relativity & Statistical Physics (University	$Z = \sum_i e^(- \beta E_i)$	`Z = \sum _i e^(-\beta E_i)`
Entropy 2.8 Quantum, Relativity & Statistical Physics (University	$S = k_B ln \Omega$	`S = k_B ln \Omega`
Canonical probability 2.8 Quantum, Relativity & Statistical Physics (University	$p_i = e^(- \beta E_i)/Z$	`p_i = e^(-\beta E_i)/Z`

4. Chemistry (64 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Moles 3.1 Mole, Stoichiometry & Solutions (Class 8	$n = mass/molarmass$	`n = mass / molar mass`
Particles 3.1 Mole, Stoichiometry & Solutions (Class 8	$N = nN_A$	`N = nN_A`
Molarity 3.1 Mole, Stoichiometry & Solutions (Class 8	$M = molesofsolute/litresofsolution$	`M = moles of solute / litres of solution`
Molality 3.1 Mole, Stoichiometry & Solutions (Class 8	$m = molesofsolute/kgofsolvent$	`m = moles of solute / kg of solvent`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Mole fraction 3.1 Mole, Stoichiometry & Solutions (Class 8	$\chi_i = n_i/\sum n_i$	`χ_i = n_i/\sum n_i`
Mass percent 3.1 Mole, Stoichiometry & Solutions (Class 8	$\%w/w = masssolute/masssolution \times 100$	`\%w/w = mass solute/mass solution \times 100`
Molar gas volume 3.1 Mole, Stoichiometry & Solutions (Class 8	$Atspecifiedconditions, V_m = V/n$	`At specified conditions, V_m = V/n`
Ideal gas 3.1 Mole, Stoichiometry & Solutions (Class 8	$PV = nRT$	`PV = nRT`
Density of ideal gas 3.1 Mole, Stoichiometry & Solutions (Class 8	$\rho = PM/RT$	`\rho = PM/RT`
Dilution 3.1 Mole, Stoichiometry & Solutions (Class 8	$M_1V_1 = M_2V_2$	`M₁V₁ = M₂V₂`
Photon energy 3.2 Atomic Structure & Nuclear Chemistry (Class 10	$E = hv = hc/\lambda$	`E = hv = hc/\lambda`
Bohr angular momentum 3.2 Atomic Structure & Nuclear Chemistry (Class 10	$mvr = nh/(2\pi) = n\hbar$	`mvr = nh/(2\pi ) = n\hbar`
Bohr radius 3.2 Atomic Structure & Nuclear Chemistry (Class 10	$r_n = a_0n^2/Z$	`r_n = a₀ n^2/Z`
Hydrogen energy 3.2 Atomic Structure & Nuclear Chemistry (Class 10	$E_n = -13.6Z^2/n^2eV$	`E_n = -13.6 Z^2/n^2 eV`
Rydberg equation 3.2 Atomic Structure & Nuclear Chemistry (Class 10	$1/\lambda = RZ^2(1/n_1^2 - 1/n_2^2)$	`1/\lambda = RZ^2(1/n₁^2-1/n₂^2)`
Radioactive decay 3.2 Atomic Structure & Nuclear Chemistry (Class 10	$N = N_0e^{(} - \lambda t)$	`N = N₀e^(-\lambda t)`
Activity 3.2 Atomic Structure & Nuclear Chemistry (Class 10	$A = \lambda N$	`A = \lambda N`
Half-life 3.2 Atomic Structure & Nuclear Chemistry (Class 10	$t_1/2 = ln2/\lambda$	`t_1/2 = ln2/\lambda`
Mean life 3.2 Atomic Structure & Nuclear Chemistry (Class 10	$\tau = 1/\lambda$	`τ = 1/\lambda`
Mass-energy 3.2 Atomic Structure & Nuclear Chemistry (Class 10	$E = \Delta mc^2$	`E = \Delta mc^2`
First law 3.3 Thermochemistry & Thermodynamics (Class 11	$\Delta U = q + w$	`\Delta U = q + w`
PV work 3.3 Thermochemistry & Thermodynamics (Class 11	$w = -P_ext\Delta V$	`w = -P_ext\Delta V`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Enthalpy 3.3 Thermochemistry & Thermodynamics (Class 11	$H = U + PV$	`H = U + PV`
Enthalpy change 3.3 Thermochemistry & Thermodynamics (Class 11	$\Delta H = \Delta U + \Delta(PV)$	`\Delta H = \Delta U + \Delta (PV)`
Gibbs energy 3.3 Thermochemistry & Thermodynamics (Class 11	$\Delta G = \Delta H - T\Delta S$	`\Delta G = \Delta H - T\Delta S`
Gibbs relation 3.3 Thermochemistry & Thermodynamics (Class 11	$\Delta G° = -RT lnK$	`\Delta G° = -RT ln K`
Entropy 3.3 Thermochemistry & Thermodynamics (Class 11	$\Delta S = q_r ev/T$	`\Delta S = q_rev/T`
Hess law 3.3 Thermochemistry & Thermodynamics (Class 11	$\Delta H_r xn = \sum \Delta H_p roducts - \sum \Delta H_r eactants$	`\Delta H_rxn = \sum \Delta H_products - \sum \Delta H_reactants`
Heat capacity 3.3 Thermochemistry & Thermodynamics (Class 11	$q = nC\Delta T$	`q = nC\Delta T`
van't Hoff 3.3 Thermochemistry & Thermodynamics (Class 11	$ln(K_2/K_1) = -\Delta H°/R(1/T_2 - 1/T_1)$	`ln(K₂/K₁)=-\Delta H°/R(1/T₂-1/T₁)`
Equilibrium constant 3.4 Chemical Equilibrium, Acids & Bases (Class 11	$K_c = [products]^c oeff/[reactants]^c oeff$	`K_c = [products]^coeff/[reactants]^coeff`
Gas equilibrium 3.4 Chemical Equilibrium, Acids & Bases (Class 11	$K_p = K_c(RT)^\Delta n$	`K_p = K_c(RT)^\Delta n`
pH 3.4 Chemical Equilibrium, Acids & Bases (Class 11	$pH = -log[H^+]$	`pH = -log[H⁺]`
pOH 3.4 Chemical Equilibrium, Acids & Bases (Class 11	$pOH = -log[OH^-]$	`pOH = -log[OH⁻]`
Water 3.4 Chemical Equilibrium, Acids & Bases (Class 11	$K_w = [H^+][OH^-]$	`K_w = [H⁺][OH⁻]`
pH-pOH 3.4 Chemical Equilibrium, Acids & Bases (Class 11	$pH + pOH = 14 at 25°C$	`pH + pOH = 14 at 25°C`
Acid dissociation 3.4 Chemical Equilibrium, Acids & Bases (Class 11	$K_a = [H^+][A^-]/[HA]$	`K_a = [H⁺][A⁻]/[HA]`
Base dissociation 3.4 Chemical Equilibrium, Acids & Bases (Class 11	$K_b = [BH^+][OH^-]/[B]$	`K_b = [BH⁺][OH⁻]/[B]`
Conjugate relation 3.4 Chemical Equilibrium, Acids & Bases (Class 11	$K_a K_b = K_w$	`K_aK_b = K_w`
Henderson–Hasselbalch 3.4 Chemical Equilibrium, Acids & Bases (Class 11	$pH = pK_a + log([A^-]/[HA])$	`pH = pK_a + log([A⁻]/[HA])`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Solubility product 3.4 Chemical Equilibrium, Acids & Bases (Class 11	$K_sp = product of ion concentrations raised to stoichiometric powers$	`K_sp = product of ion concentrations raised to stoichiometric powers`
Cell potential 3.5 Electrochemistry & Kinetics (Class 11	$E°cell = E°cathode - E°anode$	`E°cell = E°cathode − E°anode`
Nernst equation 3.5 Electrochemistry & Kinetics (Class 11	$E = E° - (RT/nF)lnQ$	`E = E° − (RT/nF)lnQ`
At 25°C 3.5 Electrochemistry & Kinetics (Class 11	$E = E° - (0.05916/n)logQ$	`E = E° − (0.05916/n)logQ`
Free energy 3.5 Electrochemistry & Kinetics (Class 11	$\Delta G = -nFE$	`\Delta G = −nFE`
Equilibrium relation 3.5 Electrochemistry & Kinetics (Class 11	$\Delta G° = -nFE°$	`\Delta G° = −nFE°`
Charge 3.5 Electrochemistry & Kinetics (Class 11	$Q = It$	`Q = It`
Faraday law 3.5 Electrochemistry & Kinetics (Class 11	$m = (MIt)/(nF)$	`m = (MIt)/(nF)`
Rate law 3.5 Electrochemistry & Kinetics (Class 11	$rate = k[A]^m[B]^n$	`rate = k[A]^m[B]^n`
First-order integrated law 3.5 Electrochemistry & Kinetics (Class 11	$ln([A]_0/[A]) = kt$	`ln([A]₀/[A]) = kt`
First-order half-life 3.5 Electrochemistry & Kinetics (Class 11	$t_1/2 = ln2/k$	`t_1/2 = ln2/k`
Arrhenius 3.5 Electrochemistry & Kinetics (Class 11	$k = Ae^(-E_a/RT)$	`k = Ae^(−E_a/RT)`
Arrhenius linear form 3.5 Electrochemistry & Kinetics (Class 11	$lnk = lnA - E_a/(RT)$	`ln k = ln A − E_a/(RT)`
Degree of unsaturation 3.6 Organic & Physical Chemistry (University	$DBE = C - H/2 + N/2 + 1 (halogens counted as H; O/S ignored)$	`DBE = C − H/2 + N/2 + 1 (halogens counted as H; O/S ignored)`
Beer–Lambert law 3.6 Organic & Physical Chemistry (University	$A = \varepsilon bc$	`A = \varepsilon bc`
Absorbance 3.6 Organic & Physical Chemistry (University	$A = log_{10}(I_0/I)$	`A = log₁₀(I₀/I)`
Raoult law 3.6 Organic & Physical Chemistry (University	$P_i = x_iP_i°$	`P_i = x_iP_i°`
Henry law 3.6 Organic & Physical Chemistry (University	$P = k_H x$	`P = k_H x`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Osmotic pressure 3.6 Organic & Physical Chemistry (University	$\pi = iCRT$	`\pi = iCRT`
Elevation of boiling point 3.6 Organic & Physical Chemistry (University	$\Delta T_b = iK_bm$	`\Delta T_b = iK_bm`
Depression of freezing point 3.6 Organic & Physical Chemistry (University	$\Delta T_f = iK_fm$	`\Delta T_f = iK_fm`
Clausius–Clapeyron 3.6 Organic & Physical Chemistry (University	$ln(P_2/P_1) = -\Delta H_vap/R(1/T_2 - 1/T_1)$	`ln(P₂/P₁)=-\Delta H_vap/R(1/T₂-1/T₁)`
Chemical potential 3.6 Organic & Physical Chemistry (University	$\mu_i = \mu_i{}^\circ + RT ln a_i$	`\mu _i = \mu _i° + RT ln a_i`
Fugacity form 3.6 Organic & Physical Chemistry (University	$\mu = \mu^\circ + RT ln(f/f^\circ)$	`\mu = \mu ° + RT ln(f/f°)`

5. Biology (24 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Magnification 4.1 Core Quantitative Biology (Class 6	$M = imagesize/objectsize$	`M = image size/object size`
Population density 4.1 Core Quantitative Biology (Class 6	$Density = Numberofindividuals/Areaorvolume$	`Density = Number of individuals/Area or volume`
Growth rate 4.1 Core Quantitative Biology (Class 6	$Growthrate = (final - initial)/time$	`Growth rate = (final-initial)/time`
Percent change 4.1 Core Quantitative Biology (Class 6	$\%change = (new - old)/old \times 100$	`\% change = (new-old)/old \times 100`
Surface area to volume 4.1 Core Quantitative Biology (Class 6	$SA : V = surfacearea/volume$	`SA:V = surface area/volume`
BMI (health metric) 4.1 Core Quantitative Biology (Class 6	$BMI = mass(kg)/height(m)^2$	`BMI = mass(kg)/height(m)^2`
Hardy–Weinberg 4.2 Genetics & Molecular Biology (Class 11	$p + q = 1$	`p + q = 1`
Hardy–Weinberg genotype 4.2 Genetics & Molecular Biology (Class 11	$p^2 + 2pq + q^2 = 1$	`p^2 + 2pq + q^2 = 1`
Allele frequency 4.2 Genetics & Molecular Biology (Class 11	$p = 2N_AA + N_Aa/2N$	`p = 2N_AA + N_Aa / 2N`
Recombination frequency 4.2 Genetics & Molecular Biology (Class 11	$RF = recombinantoffspring/totaloffspring \times 100\%$	`RF = recombinant offspring/total offspring \times 100\%`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
DNA base pairing 4.2 Genetics & Molecular Biology (Class 11	$A = T; G \equiv C$	`A=T; G≡C`
GC fraction 4.2 Genetics & Molecular Biology (Class 11	$GC\% = (G+C)/totalbases \times 100$	`GC\% = (G+C)/total bases \times 100`
PCR amplification ideal 4.2 Genetics & Molecular Biology (Class 11	$Copies \approx initialcopies \times 2^cycles$	`Copies \approx initial copies \times 2^cycles`
Michaelis–Menten 4.2 Genetics & Molecular Biology (Class 11	$v = V_max[S]/(K_m + [S])$	`v = V_max[S]/(K_m+[S])`
Catalytic efficiency 4.2 Genetics & Molecular Biology (Class 11	k_cat/K_m	`k_cat/K_m`
Hill equation 4.2 Genetics & Molecular Biology (Class 11	$\theta = [L]^n/(K_d + [L]^n)$	`\theta = [L]^n/(K_d + [L]^n)`
Exponential growth 4.3 Biostatistics & Population Biology (University	$dN/dt = rN$	`dN/dt = rN`
Exponential solution 4.3 Biostatistics & Population Biology (University	$N_t = N_0e^(rt)$	`N_t = N₀e^(rt)`
Logistic growth 4.3 Biostatistics & Population Biology (University	$dN/dt = rN(1 - N/K)$	`dN/dt = rN(1-N/K)`
Logistic solution 4.3 Biostatistics & Population Biology (University	$N(t) = K/[1 + ((K - N_0)/N_0)e^(- rt)]$	`N(t)=K/[1+((K-N₀)/N₀)e^(-rt)]`
Basic reproduction number 4.3 Biostatistics & Population Biology (University	$R_0 = expectedsecondarycasesfromoneprimarycaseinafullysusceptiblepopulation$	`R₀ = expected secondary cases from one primary case in a fully susceptible population`
Relative risk 4.3 Biostatistics & Population Biology (University	$RR = risk_exposed/risk_unexposed$	`RR = risk_exposed/risk_unexposed`
Odds ratio 4.3 Biostatistics & Population Biology (University	$OR = (ad/bc)fora2 \times 2table$	`OR = (ad/bc) for a 2 \times 2 table`
Hazard ratio 4.3 Biostatistics & Population Biology (University	$HR = hazard(t \mid group1)/hazard(t \mid group2)$	`HR = hazard(t|group1)/hazard(t|group2)`

6. Statistics & Data Science (21 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Arithmetic mean 5.1 Descriptive Statistics (Class 9	$\bar{x} = \sum x_i/n$	`\bar{x} = \sum x_i/n`
Median 5.1 Descriptive Statistics (Class 9	$Middleorderedvalue(ormeanoftwomiddlevalues)$	`Middle ordered value (or mean of two middle values)`
Range 5.1 Descriptive Statistics (Class 9	$Range = max - min$	`Range = max-min`
Population variance 5.1 Descriptive Statistics (Class 9	$\sigma^2 = \sum(x_i - \mu)^2/N$	`\sigma ^2 = \sum (x_i-\mu )^2/N`
Sample variance 5.1 Descriptive Statistics (Class 9	$s^2 = \sum(x_i - \bar{x})^2/(n-1)$	`s^2 = \sum (x_i-\bar{x})^2/(n-1)`
Standard deviation 5.1 Descriptive Statistics (Class 9	$s = textsqrts^2$	`s = \\text{sqrts}^2`
Coefficient of variation 5.1 Descriptive Statistics (Class 9	$CV = s/\bar{x} \times 100\%$	`CV = s/\bar{x} \times 100\%`
Standard error of mean 5.2 Inference & Regression (University	$SE(textbarx) = sigma/textsqrtn(ors/textsqrtn)$	`SE(\\text{bar}{x}) = \\sigma /\\text{sqrtn} (or s/\\text{sqrtn})`
z statistic 5.2 Inference & Regression (University	$z = (textbarx - \mu_0)/(sigma/textsqrtn)$	`z = (\\text{bar}{x}-\mu ₒ)/(\\sigma /\\text{sqrtn})`
t statistic 5.2 Inference & Regression (University	$t = (textbarx - \mu_0)/(s/textsqrtn)$	`t = (\\text{bar}{x}-\mu ₒ)/(s/\\text{sqrtn})`
Confidence interval 5.2 Inference & Regression (University	$estimate \pm criticalvalue \times SE$	`estimate \pm critical value \times SE`
One-proportion z 5.2 Inference & Regression (University	$z = (\hat{p} - p_0)/sqrt[p_0(1-p_0)/n]$	`z=(p̂-pₒ)/\\sqrt[pₒ(1-pₒ)/n]`
Pearson correlation 5.2 Inference & Regression (University	$r = sum[(x - textbarx)(y - textbary)]/sqrt[sum(x - textbarx)^2 sum(y - textbary)^2]$	`r = \\sum [(x-\\text{bar}{x})(y-\\text{bar}{y})]/\\sqrt[\\sum (x-\\text{bar}{x})^2 \\sum (y-\\text{bar}{y})^2]`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Simple regression 5.2 Inference & Regression (University	$\hat{y} = \beta_0 + \beta_1 x$	`\hat{y} = \beta ₀ + \beta ₁x`
Slope 5.2 Inference & Regression (University	$\beta_1 = Cov(X, Y)/Var(X)$	`\beta ₁ = Cov(X,Y)/Var(X)`
R-squared 5.2 Inference & Regression (University	$R^2 = 1 - SSE/SST$	`R^2 = 1 - SSE/SST`
Logistic regression 5.2 Inference & Regression (University	$log[p/(1-p)] = \beta_0 + \beta^T x$	`log[p/(1-p)] = \beta ₀ + \beta ᵀx`
MLE 5.2 Inference & Regression (University	\text{theta ˆ= argmax_theta L(theta |data)}	`\text{theta ˆ= argmax_theta L(theta |data)}`
Bayes posterior 5.2 Inference & Regression (University	$p(\theta \vert D) \propto p(D \vert \theta)p(\theta)$	`p(\theta |D) ∝ p(D|\theta )p(\theta )`
AIC 5.2 Inference & Regression (University	$AIC = 2k - 2ln(\hat{L})$	`AIC = 2k - 2ln(L̂)`
BIC 5.2 Inference & Regression (University	$BIC = kln(n) - 2ln(\hat{L})$	`BIC = k ln(n) - 2ln(L̂)`

7. Computer Science (23 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Binary to decimal 6.1 Algorithms & Discrete Math (Class 6	$N = \sum b_i 2^i$	`N = \sum b_i 2^i`
Arithmetic series 6.1 Algorithms & Discrete Math (Class 6	$S_n = n(a + l)/2$	`S_n = n(a+l)/2`
Geometric series 6.1 Algorithms & Discrete Math (Class 6	$S_n = a(r^n - 1)/(r - 1)$	`S_n = a(r^n-1)/(r-1)`
Graph handshaking 6.1 Algorithms & Discrete Math (Class 6	$\sum deg(v) = 2 \vert E \vert$	`\sum deg(v) = 2|E|`
Complete graph edges 6.1 Algorithms & Discrete Math (Class 6	$\vert E(K_n) \vert = n(n-1)/2$	`|E(K_n)| = n(n-1)/2`
Tree edges 6.1 Algorithms & Discrete Math (Class 6	$\vert E \vert = \vert V \vert - 1$	`|E| = |V|-1`
Binary tree max nodes 6.1 Algorithms & Discrete Math (Class 6	$Maxnodesatdepthd = 2^d$	`Max nodes at depth d = 2^d`
Binary tree max nodes 6.1 Algorithms & Discrete Math (Class 6	$Maxnodeswithheighth(rootheight0) = 2^(h + 1) - 1$	`Max nodes with height h (root height 0) = 2^(h+1)-1`
Binary search 6.2 Complexity, Information & ML (University	$T(n) = T(n/2) + O(1) = O(logn)$	`T(n)=T(n/2)+O(1) = O(log n)`
Merge sort 6.2 Complexity, Information & ML (University	$T(n) = 2T(n/2) + O(n) = O(nlogn)$	`T(n)=2T(n/2)+O(n) = O(n log n)`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Quick sort average 6.2 Complexity, Information & ML (University	$O(nlogn)average; O(n^2)worstcase$	`O(n log n) average; O(n^2) worst case`
Hash table average 6.2 Complexity, Information & ML (University	$Search/insert/delete \approx O(1)averageundergoodhashing$	`Search/insert/delete \approx O(1) average under good hashing`
Entropy 6.2 Complexity, Information & ML (University	$H(X) = -\sum p(x)log_2p(x)$	`H(X)=-\sum p(x)log₂p(x)`
Joint entropy 6.2 Complexity, Information & ML (University	$H(X,Y) = -\sum p(x,y)log_2p(x,y)$	`H(X,Y)=-\sum p(x,y)log₂p(x,y)`
Mutual information 6.2 Complexity, Information & ML (University	$I(X;Y) = \sum p(x,y)log[p(x,y)/(p(x)p(y))]$	`I(X;Y)=\sum p(x,y)log[p(x,y)/(p(x)p(y))]`
Cross entropy 6.2 Complexity, Information & ML (University	$H(p,q) = -\sum p(x)logq(x)$	`H(p,q)=-\sum p(x)log q(x)`
KL divergence 6.2 Complexity, Information & ML (University	$D_KL(p\Vert q) = \sum p(x)log[p(x)/q(x)]$	`D_KL(p||q)=\sum p(x)log[p(x)/q(x)]`
Linear regression loss 6.2 Complexity, Information & ML (University	$MSE = (1/n)\sum(y_i - \hat{y}_i)^2$	`MSE=(1/n)\sum (y_i-\hat{y}_i)^2`
Gradient descent 6.2 Complexity, Information & ML (University	$\theta_{t+1} = \theta_t - \eta\nabla J(\theta_t)$	`\theta _{t+1}=\theta _t-η\nabla J(\theta _t)`
Softmax 6.2 Complexity, Information & ML (University	$softmax(z_i) = e^{z_i}/\sum_j e^{z_j}$	`softmax(z_i)=e^{z_i}/\sum _j e^{z_j}`
Sigmoid 6.2 Complexity, Information & ML (University	$\sigma(z) = 1/(1+e^(-z))$	`\sigma (z)=1/(1+e^(-z))`
ReLU 6.2 Complexity, Information & ML (University	$ReLU(x) = max(0,x)$	`ReLU(x)=max(0,x)`
Attention 6.2 Complexity, Information & ML (University	$\text{Attention}(Q,K,V) = \text{softmax}(QK^T / textsqrtd_k)V$	`\text{Attention}(Q,K,V)=\text{softmax}(QKᵀ/\\text{sqrtd}_k)V`

8. Engineering (29 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Kirchhoff current law 7.1 Electrical & Electronics (University)	$\sum I = 0atanode$	`\sum I = 0 at a node`
Kirchhoff voltage law 7.1 Electrical & Electronics (University)	$\sum V = 0aroundaloop$	`\sum V = 0 around a loop`
RMS sinusoid 7.1 Electrical & Electronics (University)	$V_rms = V_peak/\sqrt{2}$	`V_rms = V_peak/\sqrt2`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
AC impedance resistor 7.1 Electrical & Electronics (University)	$Z_R = R$	`Z_R = R`
AC impedance inductor 7.1 Electrical & Electronics (University)	$Z_L = j\omega L$	`Z_L = j\omega L`
AC impedance capacitor 7.1 Electrical & Electronics (University)	$Z_C = 1/(j\omega C)$	`Z_C = 1/(j\omega C)`
Series RLC 7.1 Electrical & Electronics (University)	$Z = R + j(\omega L - 1/(\omega C))$	`Z = R + j(\omega L-1/(\omega C))`
Resonance 7.1 Electrical & Electronics (University)	$\omega_0 = 1/\sqrt{(LC)}$	`\omega ₒ = 1/\sqrt(LC)`
Transformer ratio 7.1 Electrical & Electronics (University)	$V_s/V_p = N_s/N_p$	`V_s/V_p = N_s/N_p`
Ideal transformer current ratio 7.1 Electrical & Electronics (University)	$I_s/I_p = N_p/N_s$	`I_s/I_p = N_p/N_s`
Power factor 7.1 Electrical & Electronics (University)	$pf = cos\varphi$	`pf = cosϕ`
Three-phase power 7.1 Electrical & Electronics (University)	$P = \sqrt{3} V_L I_L cos\varphi$	`P = \sqrt3 V_L I_L cosϕ`
Capacitor reactance 7.1 Electrical & Electronics (University)	$X_C = 1/(\omega C)$	`X_C = 1/(\omega C)`
Inductor reactance 7.1 Electrical & Electronics (University)	$X_L = \omega L$	`X_L = \omega L`
Normal stress 7.2 Mechanical & Civil Engineering (University)	$\sigma = F/A$	`\sigma = F/A`
Normal strain 7.2 Mechanical & Civil Engineering (University)	$\varepsilon = \Delta L/L$	`\varepsilon = \Delta L/L`
Hooke law 7.2 Mechanical & Civil Engineering (University)	$\sigma = E\varepsilon$	`\sigma = E\varepsilon`
Shear stress 7.2 Mechanical & Civil Engineering (University)	$\tau = V/A$	`τ = V/A`
Bending stress 7.2 Mechanical & Civil Engineering (University)	$\sigma = My/I$	`\sigma = My/I`
Torsion 7.2 Mechanical & Civil Engineering (University)	$\tau = Tr/J$	`τ = Tr/J`
Beam curvature 7.2 Mechanical & Civil Engineering (University)	$M/EI = 1/R$	`M/EI = 1/R`
Euler buckling 7.2 Mechanical & Civil Engineering (University)	$P_c r = \pi^2 EI/(KL)^2$	`P_cr = \pi ^2EI/(KL)^2`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Factor of safety 7.2 Mechanical & Civil Engineering (University)	$FoS = Failureload/Workingload$	`FoS = Failure load/Working load`
Fluid Reynolds number 7.2 Mechanical & Civil Engineering (University)	$Re = \rho vD/\mu = vD/v$	`Re = \rho vD/\mu = vD/v`
Froude number 7.2 Mechanical & Civil Engineering (University)	$Fr = v/\sqrt(gL)$	`Fr = v/\sqrt(gL)`
Mach number 7.2 Mechanical & Civil Engineering (University)	$Ma = v/a$	`Ma = v/a`
Hydraulic power 7.2 Mechanical & Civil Engineering (University)	$P = \rho gQH$	`P = \rho gQH`
Continuity 7.2 Mechanical & Civil Engineering (University)	$Q = Av$	`Q = Av`
Darcy–Weisbach 7.2 Mechanical & Civil Engineering (University)	$h_f = f(L/D)(v^2/(2g))$	`h_f = f(L/D)(v^2/(2g))`

9. Economics & Finance (27 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Compound amount 8.1 Core Finance (Class 9	$A = P(1+r)^n$	`A = P(1+r)^n`
Present value 8.1 Core Finance (Class 9	$PV = FV/(1+r)^n$	`PV = FV/(1+r)^n`
Future value 8.1 Core Finance (Class 9	$FV = PV(1+r)^n$	`FV = PV(1+r)^n`
Simple annuity PV 8.1 Core Finance (Class 9	$PV = PMT[1-(1+r)^{-}n]/r$	`PV = PMT[1-(1+r)^-n]/r`
Annuity FV 8.1 Core Finance (Class 9	$FV = PMT[(1+r)^n - 1]/r$	`FV = PMT[(1+r)^n-1]/r`
Real interest approximation 8.1 Core Finance (Class 9	$Realrate \approx Nominalrate - Inflationrate$	`Real rate \approx Nominal rate - Inflation rate`
Percentage change 8.1 Core Finance (Class 9	$\%\Delta = (new - old)/old \times 100\%$	`\%\Delta = (new-old)/old \times 100\%`
GDP expenditure 8.2 Macroeconomics & Quantitative Economics (University	$Y = C + I + G + NX$	`Y = C + I + G + NX`
National income identity 8.2 Macroeconomics & Quantitative Economics (University	$Y = C + S + T$	`Y = C + S + T`
Money identity 8.2 Macroeconomics & Quantitative Economics (University	$MV = PY$	`MV = PY`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Inflation rate 8.2 Macroeconomics & Quantitative Economics (University	$\pi = (P_t - P_{t-1})/P_{t-1}$	`\pi = (P_t-P_{t-1})/P_{t-1}`
Elasticity 8.2 Macroeconomics & Quantitative Economics (University	$\varepsilon = (\%\Delta Q)/(\%\Delta P)$	`\varepsilon = (\%\Delta Q)/(\%\Delta P)`
Price elasticity point 8.2 Macroeconomics & Quantitative Economics (University	$\varepsilon = (dQ/dP)(P/Q)$	`\varepsilon = (dQ/dP)(P/Q)`
Marginal cost 8.2 Macroeconomics & Quantitative Economics (University	$MC = dTC/dQ$	`MC = dTC/dQ`
Marginal revenue 8.2 Macroeconomics & Quantitative Economics (University	$MR = dTR/dQ$	`MR = dTR/dQ`
Profit 8.2 Macroeconomics & Quantitative Economics (University	$\pi = TR - TC$	`\pi = TR-TC`
Profit maximization 8.2 Macroeconomics & Quantitative Economics (University	$MR = MC$	`MR = MC`
Cobb–Douglas 8.2 Macroeconomics & Quantitative Economics (University	$Y = AK^{\alpha}L^{(}1 - \alpha)$	`Y = AK^\alpha L^(1-\alpha )`
Solow capital accumulation 8.2 Macroeconomics & Quantitative Economics (University	$k = sf(k) - (n + g + \delta)k$	`k = sf(k) - (n+g+\delta )k`
IS identity 8.2 Macroeconomics & Quantitative Economics (University	$Y = C(Y - T) + I(r) + G + NX$	`Y = C(Y-T)+I(r)+G+NX`
CAPM 8.2 Macroeconomics & Quantitative Economics (University	$E[R_i] = R_f + \beta_i(E[R_m] - R_f)$	`E[R_i]=R_f+\beta _i(E[R_m]-R_f)`
Portfolio return 8.2 Macroeconomics & Quantitative Economics (University	$E[R_p] = \sum w_i E[R_i]$	`E[R_p]=\sum w_iE[R_i]`
Portfolio variance 8.2 Macroeconomics & Quantitative Economics (University	$\sigma_p^2 = w^T \sum w$	`\sigma _p^2 = wᵀ\sum w`
Sharpe ratio 8.2 Macroeconomics & Quantitative Economics (University	$S = (E[R_p] - R_f)/\sigma_p$	`S=(E[R_p]-R_f)/\sigma _p`
Black–Scholes call 8.2 Macroeconomics & Quantitative Economics (University	$C = SN(d_1) - Ke^{(} - rT)N(d_2)$	`C = SN(d₁) - Ke^(-rT)N(d₂)`
d1 8.2 Macroeconomics & Quantitative Economics (University	$d_1 = [ln(S/K) + (r + sigma^2/2)T]/(sigma textsqrtT)$	`d₁=[ln(S/K)+(r+\\sigma ^2/2)T]/(\\sigma \\text{sqrtT})`
d2 8.2 Macroeconomics & Quantitative Economics (University	$d_2 = d_1 - sigma textsqrtT$	`d₂=d₁-\\sigma \\text{sqrtT}`

10. Earth & Environmental Science (15 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Pressure 9.1 Core Quantitative Earth Science (Class 6	$P = F/A$	`P = F/A`
Relative humidity 9.1 Core Quantitative Earth Science (Class 6	$RH = actualvapourpressure/saturationvapourpressure \times 100\%$	`RH = actual vapour pressure/saturation vapour pressure \times 100\%`
Hydrological balance 9.1 Core Quantitative Earth Science (Class 6	$P = ET + R + \Delta S$	`P = ET + R + \Delta S`
Population growth 9.1 Core Quantitative Earth Science (Class 6	$Growthrate = births + immigration - deaths - emigration$	`Growth rate = births + immigration - deaths - emigration`
Carbon concentration conversion 9.1 Core Quantitative Earth Science (Class 6	$ppm = partsofcomponentpermillionpartsofmixture$	`ppm = parts of component per million parts of mixture`
Heat diffusion 9.2 Geophysics & Climate (University	$\partial T/\partial t = \alpha\nabla^2 T$	`\partial T/\partial t = \alpha \nabla ^2T`
Advection-diffusion 9.2 Geophysics & Climate (University	$\partial C/\partial t + u\cdot\nabla C = D\nabla^2 C + R$	`\partial C/\partial t + u·\nabla C = D\nabla ^2C + R`
Radiative flux 9.2 Geophysics & Climate (University	$F = \sigma T^4$	`F = \sigma T^4`
Stefan–Boltzmann 9.2 Geophysics & Climate (University	$P = \varepsilon \sigma A T^4$	`P = \varepsilon \sigma AT^4`
Wien displacement 9.2 Geophysics & Climate (University	$\lambda_m ax T = b$	`\lambda _max T = b`
Hydraulic gradient 9.2 Geophysics & Climate (University	$i = \Delta h/L$	`i = \Delta h/L`
Darcy law 9.2 Geophysics & Climate (University	$q = -K\nabla h$	`q = -K\nabla h`
Richards equation 9.2 Geophysics & Climate (University	$\partial\theta/\partial t = \nabla\cdot[K(\theta)\nabla(h+z)]$	`\partial \theta /\partial t = \nabla ·[K(\theta )\nabla (h+z)]`
Seismic wave speed P 9.2 Geophysics & Climate (University	$v_P = sqrt[(K + 4\mu/3)/textrho]$	`v_P = \\sqrt[(K + 4\mu /3)/\\text{rho} ]`
Seismic wave speed S 9.2 Geophysics & Climate (University	$v_S = \sqrt{(\mu/\rho)}$	`v_S = \sqrt(\mu /\rho )`

11. Health Sciences (9 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Cardiac output 10.1 Core Quantitative & Physiological Relations (University)	$CO = HR \times SV$	`CO = HR \times SV`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Mean arterial pressure approximation 10.1 Core Quantitative & Physiological Relations (University)	$MAP \approx DBP + (SBP-DBP)/3$	`MAP \approx DBP + (SBP-DBP)/3`
Body mass index 10.1 Core Quantitative & Physiological Relations (University)	$BMI = mass(kg)/height(m)^2$	`BMI = mass(kg)/height(m)^2`
Oxygen delivery 10.1 Core Quantitative & Physiological Relations (University)	$D_O2 = CO \times C_aO2 \times constant$	`D_O2 = CO \times C_aO2 \times constant`
Alveolar ventilation 10.1 Core Quantitative & Physiological Relations (University)	$V_A = (V_T-V_D)f$	`V_A = (V_T-V_D)f`
Clearance 10.1 Core Quantitative & Physiological Relations (University)	$CL = Rate of elimination / Plasma concentration$	`CL = Rate of elimination / Plasma concentration`
Half-life (first order) 10.1 Core Quantitative & Physiological Relations (University)	$t_1/2 = 0.693/k$	`t_1/2 = 0.693/k`
Drug infusion steady state 10.1 Core Quantitative & Physiological Relations (University)	$C_ss = Rate_in/CL$	`C_ss = Rate_in/CL`
Exponential elimination 10.1 Core Quantitative & Physiological Relations (University)	$C(t)=C_0e^(-kt)$	`C(t)=C₀e^(-kt)`

12. Psychology & Social Science (10 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Standard score 11.1 Quantitative Methods (University	$z=(x-\mu)/\sigma$	`z=(x-\mu )/\sigma`
Correlation 11.1 Quantitative Methods (University	$r = Cov(X,Y)/(\sigma _X\sigma _Y)$	`r = Cov(X,Y)/(\sigma _X\sigma _Y)`
Linear model 11.1 Quantitative Methods (University	$Y = \beta_0 + \beta_1X + \varepsilon$	`Y = \beta ₀ + \beta ₁X + \varepsilon`
Multiple regression 11.1 Quantitative Methods (University	$y = X\beta + \varepsilon$	`y = X\beta + \varepsilon`
Coefficient of determination 11.1 Quantitative Methods (University	$R^2 = 1-SSE/SST$	`R^2 = 1-SSE/SST`
Odds ratio 11.1 Quantitative Methods (University	$OR = odds_exposed/odds_unexposed$	`OR = odds_exposed/odds_unexposed`
Cronbach's alpha 11.1 Quantitative Methods (University	$\alpha = k/(k-1)[1-\sum \sigma _i^2/\sigma_total^2]$	`\alpha = k/(k-1)[1-\sum \sigma _i^2/\sigma _total^2]`
Cohen's d 11.1 Quantitative Methods (University	$d=(\bar{x}_1-\bar{x}_2)/s_pooled$	`d=(\bar{x}₁-\bar{x}₂)/s_pooled`
Pooled SD 11.1 Quantitative Methods (University	$s_p = sqrt[((n_1-1)s_1^2+(n_2-1)s_2^2)/(n_1+n_2-2)]$	`s_p = \\sqrt[((n₁-1)s₁^2+(n₂-1)s₂^2)/(n₁+n₂-2)]`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Standard error 11.1 Quantitative Methods (University	$SE = SD/$ $textsqrtn$	`SE = SD/\\text{sqrtn}`

13. Business, Accounting & Operations (12 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Accounting equation 12.1 Accounting & Operations (University)	$Assets = Liabilities + Equity$	`Assets = Liabilities + Equity`
Gross profit 12.1 Accounting & Operations (University)	$Grossprofit = Revenue - Costofgoodssold$	`Gross profit = Revenue - Cost of goods sold`
Operating profit 12.1 Accounting & Operations (University)	$EBIT = Revenue - Operatingexpenses$	`EBIT = Revenue - Operating expenses`
Net profit 12.1 Accounting & Operations (University)	$Netprofit = Revenue - Totalexpenses$	`Net profit = Revenue - Total expenses`
Current ratio 12.1 Accounting & Operations (University)	$Currentratio = Currentassets/Currentliabilities$	`Current ratio = Current assets/Current liabilities`
Quick ratio 12.1 Accounting & Operations (University)	$Quickratio = Quickassets/Currentliabilities$	`Quick ratio = Quick assets/Current liabilities`
Break-even units 12.1 Accounting & Operations (University)	$BEunits = Fixedcosts/(Sellingprice - Variablecostperunit)$	`BE units = Fixed costs/(Selling price-Variable cost per unit)`
Contribution 12.1 Accounting & Operations (University)	$Contribution = Sales - Variablecosts$	`Contribution = Sales - Variable costs`
Contribution margin ratio 12.1 Accounting & Operations (University)	$CMratio = Contribution/Sales$	`CM ratio = Contribution/Sales`
EOQ 12.1 Accounting & Operations (University)	$EOQ = \sqrt(2DS/H)$	`EOQ = \sqrt(2DS/H)`
Inventory turnover 12.1 Accounting & Operations (University)	$Inventoryturnover = COGS/Averageinventory$	`Inventory turnover = COGS/Average inventory`
ROI 12.1 Accounting & Operations (University)	$ROI = Gain/Investment \times 100\%$	`ROI = Gain/Investment \times 100\%`

14. Astronomy & Astrophysics (10 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Inverse square flux 13.1 Core Formulae (University	$F = L/(4\pi r^2)$	`F = L/(4\pi r^2)`
Luminosity 13.1 Core Formulae (University	$L = 4\pi R^2\sigma T_eff^4$	`L = 4\pi R^2\sigma T_eff^4`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Magnitude difference 13.1 Core Formulae (University	$m_1 - m_2 = -2.5 log_{10}(F_1/F_2)$	`m₁−m₂ = −2.5 log₁₀(F₁/F₂)`
Distance modulus 13.1 Core Formulae (University	$m - M = 5log_{10}(d/10pc)$	`m−M = 5log₁₀(d/10 pc)`
Kepler third law 13.1 Core Formulae (University	$P^2 = 4\pi^2a^3/[G(M+m)]$	`P^2 = 4\pi ^2a^3/[G(M+m)]`
Schwarzschild radius 13.1 Core Formulae (University	$r_s = 2GM/c^2$	`r_s = 2GM/c^2`
Hubble law 13.1 Core Formulae (University	$v = H_0d$	`v = H₀d`
Critical density 13.1 Core Formulae (University	$\rho_c = 3H^2/(8\pi G)$	`\rho _c = 3H^2/(8\pi G)`
Friedmann equation 13.1 Core Formulae (University	$H^2 = (8\pi G/3)\rho - kc^2/a^2 + \Lambda c^2/3$	`H^2 = (8\pi G/3)\rho − kc^2/a^2 + Λc^2/3`
Redshift 13.1 Core Formulae (University	$1 + z = \lambda_obs/\lambda_emit$	`1+z = \lambda _obs/\lambda _emit`

15. Reference (14 Formulas)

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Pi 15.1 Common Mathematical & Physical Constants (All Levels)	$\pi \approx 3.141592653589793$	`\pi \approx 3.141592653589793`
Euler number 15.1 Common Mathematical & Physical Constants (All Levels)	$e \approx 2.718281828459045$	`e \approx 2.718281828459045`
Speed of light 15.1 Common Mathematical & Physical Constants (All Levels)	$c = 299,792,458m/s$	`c = 299,792,458 m/s`
Gravitational constant 15.1 Common Mathematical & Physical Constants (All Levels)	$G \approx 6.67430 \times 10^-11m^3kg^-1s^-2$	`G \approx 6.67430\times 10^−11 m^3 kg^−1 s^−2`
Planck constant 15.1 Common Mathematical & Physical Constants (All Levels)	$h \approx 6.62607015 \times 10^-34J \cdot s$	`h \approx 6.62607015\times 10^−34 J·s`
Reduced Planck constant 15.1 Common Mathematical & Physical Constants (All Levels)	$\hbar = h/(2\pi)$	`\hbar = h/(2\pi )`
Elementary charge 15.1 Common Mathematical & Physical Constants (All Levels)	$e \approx 1.602176634 \times 10^-19C$	`e \approx 1.602176634\times 10^−19 C`
Vacuum permittivity 15.1 Common Mathematical & Physical Constants (All Levels)	$\varepsilon_0 \approx 8.8541878128 \times 10^-12F/m$	`\varepsilon ₀ \approx 8.8541878128\times 10^−12 F/m`
Vacuum permeability 15.1 Common Mathematical & Physical Constants (All Levels)	$\mu_0 \approx 1.25663706212 \times 10^-6H/mapproximately$	`\mu ₀ \approx 1.25663706212\times 10^−6 H/m approximately`
Boltzmann constant 15.1 Common Mathematical & Physical Constants (All Levels)	$k_B \approx 1.380649 \times 10^-23J/K$	`k_B \approx 1.380649\times 10^−23 J/K`

CONCEPT / FORMULA NAME	RENDERED MATHEMATICAL OUTPUT	LATEX CODE
Avogadro constant 15.1 Common Mathematical & Physical Constants (All Levels)	$N_A = 6.02214076\times 10^23 mol^-1$	`N_A = 6.02214076\times 10^23 mol^-1`
Gas constant 15.1 Common Mathematical & Physical Constants (All Levels)	$R = 8.314462618 J mol^-1 K^-1$	`R = 8.314462618 J mol^-1 K^-1`
Faraday constant 15.1 Common Mathematical & Physical Constants (All Levels)	$F \approx 96485.33212 C/mol$	`F \approx 96485.33212 C/mol`
Standard gravity 15.1 Common Mathematical & Physical Constants (All Levels)	$g_0 = 9.80665 m/s^2$	`g₀ = 9.80665 m/s^2`

16. Mathematical Operators, Relations, Sets, Logic & Arrows

Symbol	LaTeX	Description
$\pm$	`\pm`	Plus-minus
$\mp$	`\mp`	Minus-plus
$\times$	`\times`	Multiplication cross
$\div$	`\div`	Division sign
$\cdot$	`\cdot`	Center dot product
$\circ$	`\circ`	Function composition
$\ast$	`\ast`	Asterisk / convolution
$\oplus$	`\oplus`	Direct sum
$\otimes$	`\otimes`	Tensor product
$\odot$	`\odot`	Hadamard product
$\dagger$	`\dagger`	Hermitian conjugate
$=$	`=`	Equals
$\neq$	`\neq`	Not equal
$\approx$	`\approx`	Approximately equal
$\equiv$	`\equiv`	Identical / congruent
$\sim$	`\sim`	Similar / distributed as
$\propto$	`\propto`	Proportional to
$<$	`<`	Less than
$>$	`>`	Greater than
$\leq$	`\leq`	Less than or equal
$\geq$	`\geq`	Greater than or equal
$\ll$	`\ll`	Much less than
$\gg$	`\gg`	Much greater than
$\cong$	`\cong`	Isomorphic / congruent
$\in$	`\in`	Element of
$\notin$	`\notin`	Not element of
$\subset$	`\subset`	Proper subset
$\subseteq$	`\subseteq`	Subset or equal
$\not\subset$	`\not\subset`	Not a subset
$\supset$	`\supset`	Superset
$\supseteq$	`\supseteq`	Superset or equal
$\cup$	`\cup`	Set union
$\cap$	`\cap`	Set intersection
$\emptyset$	`\emptyset`	Empty set
$\setminus$	`\setminus`	Set difference
$\forall$	`\forall`	For all
$\exists$	`\exists`	There exists
$\nexists$	`\nexists`	There does not exist
$\neg$	`\neg`	Logical NOT
$\land$	`\land`	Logical AND
$\lor$	`\lor`	Logical OR
$\therefore$	`\therefore`	Therefore
$\because$	`\because`	Because / since
$\mathbb{R}$	`\mathbb{R}`	Real numbers
$\mathbb{C}$	`\mathbb{C}`	Complex numbers
$\mathbb{N}$	`\mathbb{N}`	Natural numbers
$\mathbb{Z}$	`\mathbb{Z}`	Integers
$\mathbb{Q}$	`\mathbb{Q}`	Rational numbers
$\rightarrow$	`\rightarrow`	Right arrow
$\leftarrow$	`\leftarrow`	Left arrow
$\leftrightarrow$	`\leftrightarrow`	Bidirectional arrow
$\Rightarrow$	`\Rightarrow`	Implies
$\Leftarrow$	`\Leftarrow`	Is implied by
$\Leftrightarrow$	`\Leftrightarrow`	If and only if (IFF)
$\mapsto$	`\mapsto`	Maps to
$\uparrow$	`\uparrow`	Up arrow
$\downarrow$	`\downarrow`	Down arrow
∞	`\infty`	Infinity
∂	`\partial`	Partial differential
∇	`\nabla`	Nabla / Del gradient
$\hbar$	`\hbar`	Reduced Planck constant
ℓ	`\ell`	Script l
$\aleph$	`\aleph`	Aleph cardinal
$\angle$	`\angle`	Angle
$\perp$	`\perp`	Perpendicular
$\parallel$	`\parallel`	Parallel
$\dots$	`\dots`	Horizontal ellipsis
$\vdots$	`\vdots`	Vertical ellipsis
$\ddots$	`\ddots`	Diagonal ellipsis

17. Complete Greek Alphabet (Lowercase & Uppercase)

α `\alpha` alpha (Lower)	β `\beta` beta (Lower)	γ `\gamma` gamma (Lower)	δ `\delta` delta (Lower)	ϵ `\epsilon` epsilon (Lower)	ε `\varepsilon` varepsilon (Lower)
ζ `\zeta` zeta (Lower)	η `\eta` eta (Lower)	θ `\theta` theta (Lower)	ϑ `\vartheta` vartheta (Lower)	ι `\iota` iota (Lower)	κ `\kappa` kappa (Lower)
λ `\lambda` lambda (Lower)	μ `\mu` mu (Lower)	ν `\nu` nu (Lower)	ξ `\xi` xi (Lower)	π `\pi` pi (Lower)	ϖ `\varpi` varpi (Lower)
ρ `\rho` rho (Lower)	ϱ `\varrho` varrho (Lower)	σ `\sigma` sigma (Lower)	ς `\varsigma` varsigma (Lower)	τ `\tau` tau (Lower)	υ `\upsilon` upsilon (Lower)
φ `\phi` phi (Lower)	ϕ `\varphi` varphi (Lower)	χ `\chi` chi (Lower)	ψ `\psi` psi (Lower)	ω `\omega` omega (Lower)	Γ `\Gamma` Gamma (Upper)
Δ `\Delta` Delta (Upper)	Θ `\Theta` Theta (Upper)	Λ `\Lambda` Lambda (Upper)	Ξ `\Xi` Xi (Upper)	Π `\Pi` Pi (Upper)	Σ `\Sigma` Sigma (Upper)
Υ `\Upsilon` Upsilon (Upper)	Φ `\Phi` Phi (Upper)	Ψ `\Psi` Psi (Upper)	Ω `\Omega` Omega (Upper)		